G H Raisoni International Skill Tech University, Pune.

Question Paper

Continuous Assessment Examination-III (CAE-III)

Programme: B. Tech | **Year: First** | **Semester: I** | **Max Marks: 20**

Duration: 60 mins. | **Course Name: Introduction to Cyber Security** | **Course Code: OT109**

Note: This question paper addresses below Course Outcomes (CO's):

CO1	Understand cyber security fundamentals, the CIA Triad, and ethical practices.
CO3	Design secure networks and evaluate encryption and key management strategies.

Q. No.	Sub Q. No.	Question	Mks
1		Solve any **TWO**	
	A	a) Explain the CIA triad for a healthcare organization. b) Develop an ethical protocol for handling a data breach in a medical institution, addressing: i) Legal compliance requirements ii) Stakeholder communication iii) Initial forensic investigation steps	05
	B	a) Rahim is a cybersecurity consultant for a small e-commerce startup. Develop a risk assessment strategy that: i) Identifies potential financial threats ii) Calculates the potential economic impact of a data breach iii) Proposes cost-effective security investments iv) Demonstrates ROI of cybersecurity measures b) Help Rahim to create an employee cyber security training program that: i) Provide training on psychological tactics used by cybercriminals ii) Demonstrates real-world consequences of digital negligence iii) Help to get employees to take immediate steps in case any attacks happen. **Scenario Context:** The startup processes 10,000 monthly transactions with sensitive customer payment information.	05
	C	a) If a software developer embeds a backdoor into their codebase to gain access to client systems for debugging purposes without informing the client or employer. What could be the ethical implications of this action under cyber security ethics? What risks does this pose to the organization and its clients? b) Comment on the recent use of hacking as cyberwarfare by different countries, if proven should it be considered as an act of crime in the international court of justice?	05
	D	Analyze the evolving cybercrime ecosystem as a) What new domains are evolving as cybersecurity threats? b) How cost factor has increased in recent years c) How defence strategies have evolved?	05
2		Solve any **TWO**	
	A	a) Give steps that can be followed to configure a firewall b) Explain Secure network design principles in detail	05
	B	a) A retail company suffered a large-scale breach because hackers accessed its payment processing network via an insecure HVAC system. Using this case study, explain the importance of network segmentation and isolation in secure network design. b) Suggest how to design a secure network for an e-commerce platform where customer data, payment systems, and employee networks must remain segregated.	05
	C	a) Explain any one Encryption algorithm with an example. b) Hybrid encryption combines the benefits of both symmetric and asymmetric techniques. Propose a real-life scenario where hybrid encryption would be the most suitable choice and **justify your reasoning**.	05
	D	a) An organization performs weekly data backups but discovers that its backups are also encrypted during a ransomware attack. What could be the flaws in their backup strategy and how a more robust approach can ensure the availability and integrity of backup data? b) What is a ransomware attack? How it happens? Give key steps of its planning and execution	05